

Decentralized electric vehicle charging enables large-scale photovoltaic integration in tropical cities

Received: 14 May 2025

Accepted: 13 March 2026

Published online: 07 April 2026

 Check for updatesJiazu Zhou^{1,2}, Tianyu Dong^{1,2}, Hongrong Yang¹ , Seanglidet Yean²,
Bu-Sung Lee^{2,3} & Markus Schläpfer¹ ✉

The rapid growth of tropical cities and the rising challenges of climate change call for efficient, low-carbon energy systems. Solar photovoltaics could play a key role, but deployment in tropical climates is constrained by localized thunderstorms that cause rapid generation fluctuations and stress electricity grids. While electric vehicles could balance such fluctuations by acting as distributed energy storage, this potential has not been systematically explored. Here, using Singapore as a case study, we develop a decentralized, district-level vehicle charging strategy that aligns with urban mobility patterns inferred from mobile phone data. Contrary to conventional centralized charging strategies, our approach substantially reduces grid flows, enabling greater photovoltaic integration into the existing grid infrastructure. We further show that detailed urban mobility patterns are critical to the balancing performance of electric vehicle storage. Our results highlight the potential of coordinated photovoltaic and electric vehicle systems for large-scale solar energy deployment in tropical cities.

Urban growth is increasingly concentrated in tropical regions where $\approx 50\%$ of the world's population is expected to live by 2050¹. This trend, particularly evident in ever-expanding megacities such as Lagos, São Paulo, or Jakarta, is associated with rapidly rising energy demands and a pressing need for low-carbon energy infrastructures that limit the impact on global warming².

Solar photovoltaic (PV) is currently becoming one of the key renewable energy technologies to meet the global growth of electricity demand while reducing carbon emissions^{3–6}. Indeed, PV installations have experienced rapid global expansion, driven by declining costs and supportive policies⁷. However, due to the intermittent nature of solar energy, the large-scale integration of PV installations poses several operational challenges to the electricity system^{8,9}. A prominent example is the so-called ‘duck curve’, where the net electricity demand (total demand minus PV output) across a region dips when PV injections become high at noon (‘duck’s belly’),

and then suddenly rises when the sun sets in the early evening while people still need high amounts of electricity (‘duck’s neck’). As a result, electricity system operators face an oversupply of generation during the valley hours and steep demand increases in the early evening that may require PV curtailments¹⁰ and large amounts of highly flexible generators like gas turbines¹¹. While the duck curve problem has so far been observed typically in more dry climates like California in the USA¹² and Shandong in China¹³, its challenges intensify in tropical climates where localized thunderstorms and rapidly changing cloud covers additionally introduce highly fluctuating PV injections that vary strongly across small geographic areas^{14,15} (Supplementary Note 1). Such periods of strong local imbalances between electricity supply and demand may jeopardize the security of the electricity supply if there are insufficient electricity grid capacities to balance out these fluctuations across larger regions^{16–18}. Therefore, addressing the intermittent availability of

¹Department of Civil Engineering and Engineering Mechanics, Columbia University, New York, NY, USA. ²Future Cities Laboratory Global, Singapore-ETH Centre, Singapore, Singapore. ³College of Computing and Data Science, Nanyang Technological University, Singapore, Singapore.

✉ e-mail: m.schlaepfer@columbia.edu

solar energy is essential to enabling the deployment of large-scale PV energy in tropical cities.

Controlled charging of the growing number of electric vehicles (EVs)^{19–21} may offer a scalable and cost-efficient solution to these challenges^{22–25}. For instance, at the city-wide aggregated power system level, previous work has shown that incentivizing vehicle charging during the daytime, when PV generation is high, flattens the duck curve, which reduces the need for PV curtailments or capital investments in flexible generators, stationary energy battery storage or other technologies that would otherwise be required to balance out the intermittent generation^{26–28}. The balancing effects are further amplified if EVs can feed the energy stored in their batteries back to the electricity grid. This bidirectional energy exchange through vehicle charging and discharging is known as vehicle-to-grid (V2G) and effectively turns EVs into large amounts of decentralized, mobile energy storage devices²⁹.

These well-studied dynamics at the aggregate power systems level^{23,26–28,30,31} suggest that controlled EV charging can also help balance out the more local and more short-term PV generation fluctuations present in tropical regions. This may not only flatten the net electricity demand curve at the system level but also reduce the need for costly grid expansions by minimizing the flows resulting from strong spatial and temporal imbalances in PV generation. However, despite the growing importance of tropical cities, this potential has not been explored systematically, possibly due to the lack of detailed individual mobility data needed to assess the spatial and temporal availability of EVs for charging and discharging.

Here, we address this gap by integrating fine-grained EV mobility patterns derived from mobile phone data and PV generation patterns derived from solar irradiance data into an electricity grid model. Using Singapore as a case study of a tropical city and applying different future EV and PV integration scenarios, our framework enables us to assess EV charging strategies for their ability to support large-scale PV integration by balancing highly intermittent PV generation in tropical climates. Our analysis reveals that the commonly assumed EV charging optimization at the system level flattens the city-wide duck curve and reduces the associated peak net demand, but – counterintuitively – induces even larger flows on the electricity grid. In contrast, controlled EV charging optimized to smooth fluctuations at the more local urban district level not only flattens the city-wide duck curve but also reduces grid flows, thereby lowering the need for infrastructure upgrades. In addition, we demonstrate the importance of considering detailed urban mobility patterns in evaluating the EV potential by comparing the effectiveness of controlled EV charging on weekdays and weekends.

Results

PV integration and EV charging scenarios

In Singapore, a tropical city-state with ≈ 5.9 million inhabitants, solar energy is considered the only viable option for large-scale renewable energy use due to limited resources otherwise³². To assume realistic future high-PV-high-EV scenarios, we follow the projections of Singapore's 'PV Roadmap'³³ and 'Green Plan'³⁴, according to which the maximum installed PV capacity reaches 8.6 gigawatt-peak (GW_p) and all cars ($N \approx 0.7 \cdot 10^6$) are assumed to be electric by the year 2050. PV panels are thereby installed on all usable surfaces, including roofs and façades of buildings, infrastructures, and suitable water bodies. For the same year, the city's peak base electricity demand (i.e., without EVs and PV) is estimated to be 11.5 GW (see Methods). Note that the number of cars per resident is low (≈ 0.12), even when compared to other 'car-lite', high-density cities (e.g., ≈ 0.23 in New York City).

Based on these values for 2050, we consider five different scenarios to quantify the benefits of controlled EV charging for the integration of PV: *i*) "No PV, no EV" – neither PV nor EVs are integrated, *ii*) "PV only" – PV is integrated but without any EV adoption, *iii*) "PV

+uncontrolled charging" – both PV and EVs are integrated and EVs simply charge until their batteries are full when they are parked, *iv*) "V2G only" – no PV is integrated but EVs are adopted and obey controlled bidirectional charging where both charging and discharging behaviors are optimized, *v*) "PV+V2G" – both PV and EVs are integrated and EVs obey controlled bidirectional charging.

We simulate the spatial and temporal variation of the PV generation by combining solar irradiance data with fine-grained building geometry information (Methods). In addition, the detailed mobility patterns of EVs are needed to determine the periods and locations when they are parked and, therefore, available for charging and discharging³⁵. To that end, we simulate the EV mobility using mobile phone data and the TimeGeo model described in refs. 31,36 (Supplementary Notes 2 and 3). This framework generates detailed daily trajectories (i.e., time-stamped sequences of visits to different locations) for each individual of an entire population by enriching sparse location data from mobile phones with a probabilistic human mobility model. Having established these general mobility patterns, we then use transportation mode data to extract the trips that are made by EVs (Supplementary Note 3) and to determine charging and discharging patterns. A series of validation studies using independent ground-truth datasets, including census data and carpark occupancy records, confirm that the mobile phone data are sufficiently representative across Singapore and that our EV mobility estimates are consistent with empirical observations (details in Supplementary Note 3). Finally, to study the combined impact of PV and EV integration on the power flows across the city-wide electricity grid, we make use of an existing electricity network model for Singapore³⁷ (Supplementary Note 5). Possible future changes in the network topology are thereby not considered.

Impact of tropical thunderstorms on grid flows

We start assessing the impact of thunderstorms by examining the dynamic grid loadings with PV integration ("PV only" scenario). Figure 1a depicts the solar irradiance at 10:30 am, 12:00 pm, and 1:30 pm of a typical day with thunderstorms. Solar irradiance shows significant variations over short spatial and temporal scales due to thunderstorms. For instance, differences of up to 1000 W/m² can be observed between the east and west part of the city (≈ 55 km distance), and local fluctuations of 800 W/m² within the short period between 10:30 am and 12:00 pm. Lower irradiance during thunderstorms leads to rapid decreases in the PV generation and therefore to sharp local peaks in the net electricity demands (Supplementary Note 6). This results in dramatically increasing grid flows that balance out these demand differences. Indeed, as can be observed by comparing Fig. 1a with Fig. 1b, districts with lower solar irradiance tend to exhibit higher line loads, and vice versa. Figure 1c additionally illustrates the detailed load profile of an exemplary line during the same day, with periods affected by thunderstorms highlighted. During the two thunderstorm periods, the line load shows sharp peaks, followed by rapid drops as the thunderstorms pass.

As shown in Fig. 1d for all lines and an extended period of 2 months (using solar irradiance data from March 20th to May 20, 2024), these significant increases in the line loads due to thunderstorms persist after integrating EVs without a controlled charging scheme ("PV+uncontrolled charging", see Methods). On several lines, maximum loads increase by more than 100% on days with thunderstorms (characterized by strong irradiance fluctuations) compared with cloud-free, high-irradiance days. This shows that even conservative security margins of 33–50% can be insufficient if line flow limits do not account for thunderstorms (a 100% load increase corresponds to a required security margin of 50%). Such increases in peak line loads therefore put high pressure on the transmission lines and, without energy storage, may require costly upgrades to the grid infrastructure.

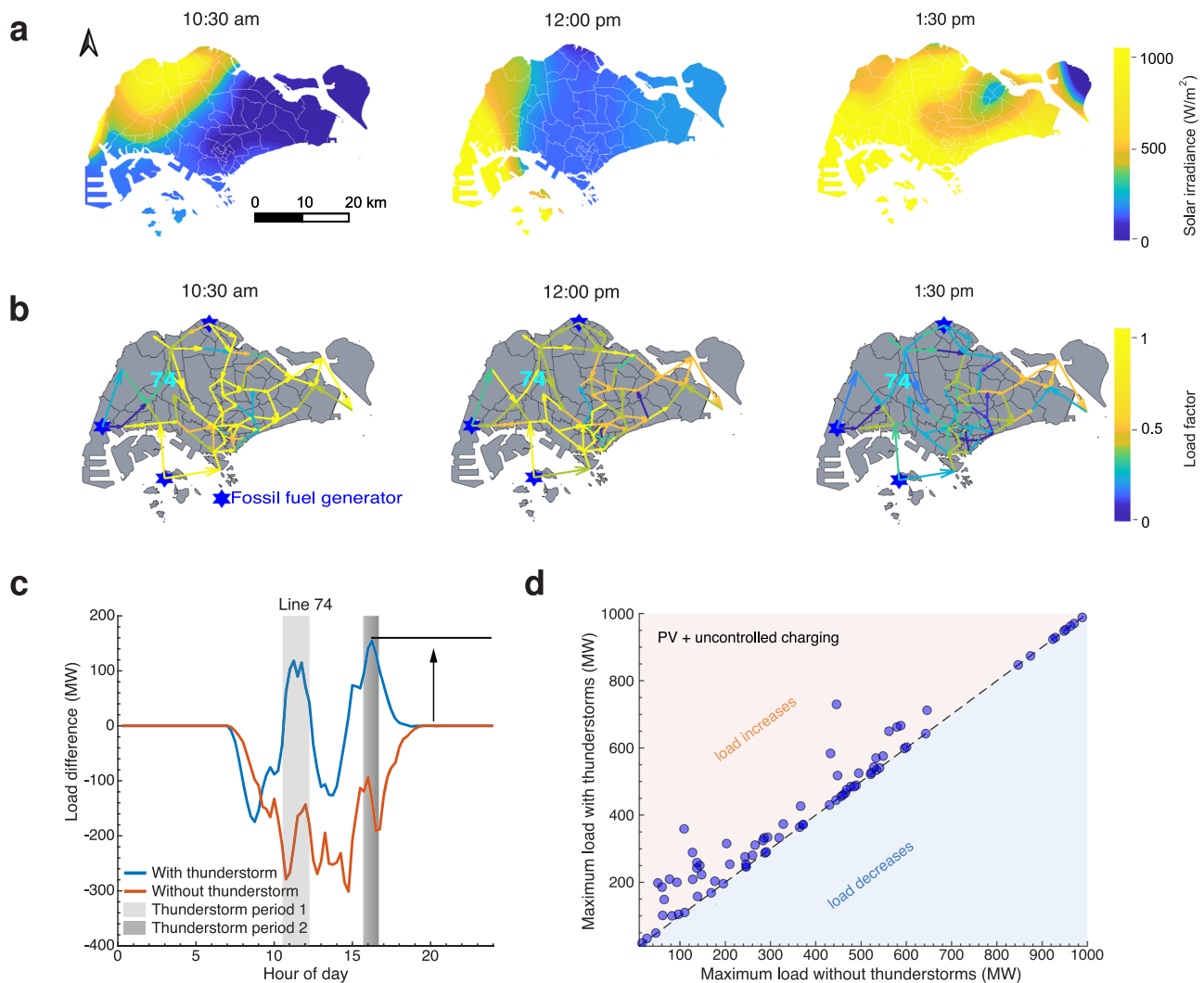


Fig. 1 | Tropical thunderstorms lead to additional stress on electricity grids. **a** Solar irradiance on a typical day with thunderstorms (April 20, 2024). The white lines are urban district borders. **b** Resulting loads of the lines expressed as the fraction of their maximum load on that day (load factor) in the "PV only" scenario. **c** Detailed temporal load variations on the exemplary line 74 (depicted in b), measured as the difference between the "PV only" scenario and the baseline load of the "No PV, no EV" scenario. To highlight the impact of thunderstorms, we also plot load variation on the same day, assuming no cloud coverage. **d** Thunderstorm-

induced load increases of all lines during a simulated 2-month period of the "PV + uncontrolled charging" scenario. For each line, the maximum load over all days with thunderstorms, versus the maximum load over all days with no cloud coverage, is plotted. For the majority of lines, the maximum load is substantially higher during days with thunderstorms (lines with a maximum load larger than 1000 MW (7 out of 85) show no effect and are omitted for visualization purposes). PV stands for photovoltaic, and EV stands for electric vehicle.

Mitigating the impact of thunderstorms through EV charging

Having established the impact of thunderstorms, we now examine the potential of controlled EV charging for mitigating these additional burdens on the electricity grid ("PV+V2G" scenario). We start with a conventional 'system-level' V2G optimization approach which seeks to minimize the aggregated, city-wide peak net demand and is typically used to assess the benefits of V2G for the power generation level^{27,28} (Methods). Figure 2a shows the effect of this controlled EV charging scheme on the flow dynamics of an exemplary line (same as in Fig. 1c) over the course of a day. Interestingly, optimizing EV charging at the system level leads to a strong peak in the flow magnitude that is not only higher than in the "PV only" scenario but even exceeds the maximum flow observed in the "PV+uncontrolled charging" scenario. This observation may appear counterintuitive, as V2G, unlike uncontrolled charging, enables vehicles to feed power back to the grid, reducing net electricity demand and, consequently, line flows. However, the system-level "PV+V2G" strategy prioritizes minimizing peak net demand at the city-wide scale, which permits

localized demand fluctuations. These local variations, in turn, lead to significant line flow variability.

To tackle this challenge, we apply a decentralized, 'district-level' V2G optimization scheme (Methods). It minimizes the peak net electricity demands of each urban district and thus can be expected to alleviate the observed thunderstorm-induced grid flows. Singapore comprises 55 urban districts ('planning areas') with areas ranging from $\approx 1 \text{ km}^2$ in the dense downtown to $\approx 20 \text{ km}^2$ in the surroundings. Indeed, district-level "PV+V2G" is able to reduce the peak load on the exemplary line compared to both the "PV only" scenario and the "PV + uncontrolled charging" scenario (Fig. 2b). This reflects its focus on flattening the demand curves at the more local level, which reduces the need for electricity exchanges across districts and thus naturally lowers the line flows. Note that these peaks are, however, still higher than in the scenarios without PV ("No PV, no EV" and "V2G only").

Extending the analysis to all lines and to the 2-month period of solar irradiance data generalizes these observations (Fig. 2c-f). On sunny days without thunderstorms, both system-level (Fig. 2c) and

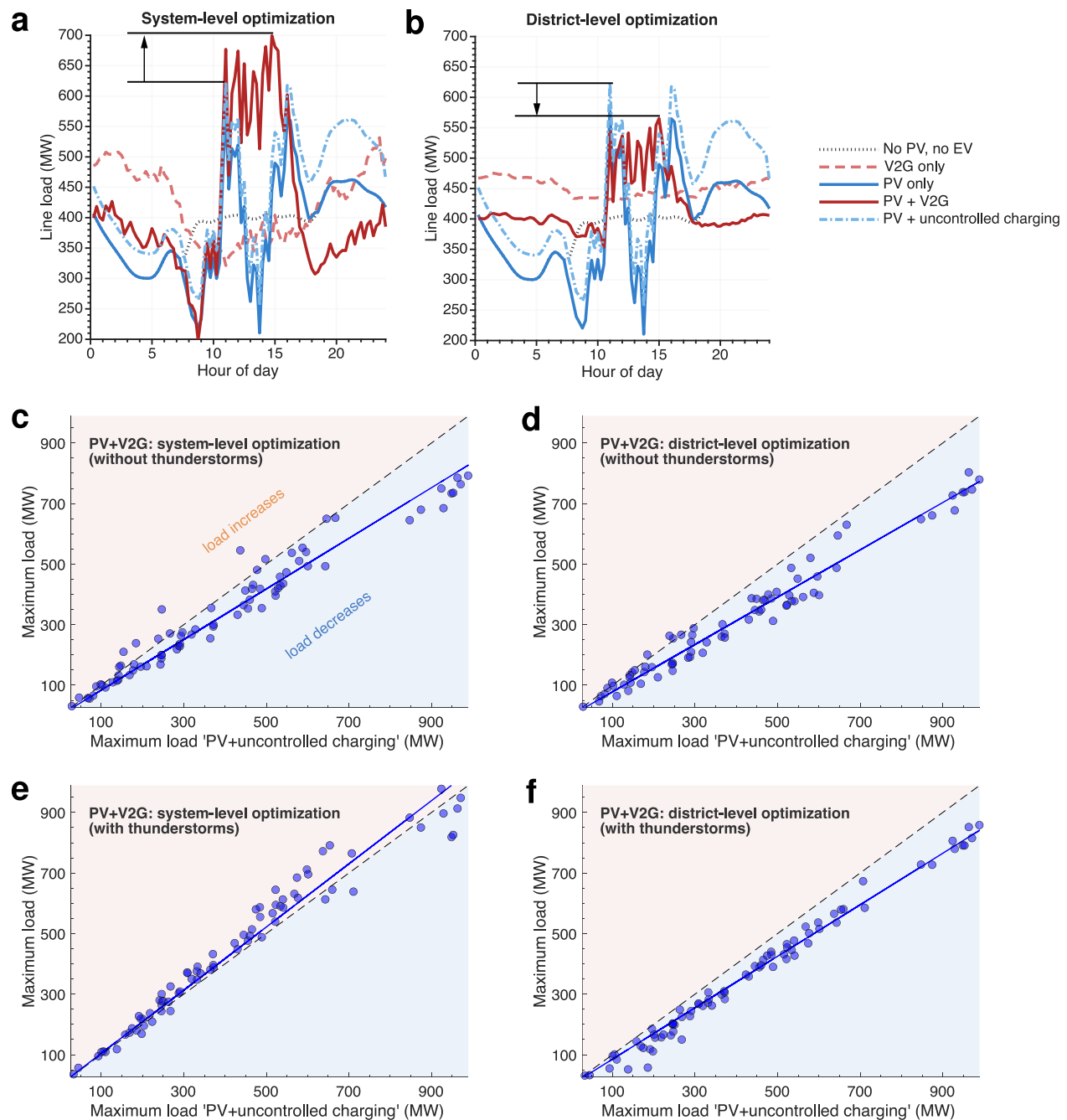


Fig. 2 | District-level electric vehicle charging reduces grid stress in tropical climates. a, b Flows on the same exemplary line as in Fig. 2c for the same day with thunderstorms when optimizing EV charging at the system level (a) and at the district level (b), compared to all other PV and EV integration scenarios (here the flows are shown in absolute values). **c–f** Effectiveness of EV charging optimization. For all sunny days (c, d) and all thunderstorm days (e, f) of the 2-month period, the maximum load of each line after system-level (c, e) and district-level (d, f)

optimization is plotted against the corresponding value in the “PV+uncontrolled charging” scenario. The continuous lines are linear fits to the data. A simulated weekday is taken for the daily mobility patterns. For enhanced visualization, lines with maximum flows of more than 1000 MW (7 out of 85) are shown separately in Supplementary Fig. S14. EV stands for electric vehicle, PV stands for photovoltaic, and V2G stands for vehicle-to-grid.

district-level (Fig. 2d) charging optimization are able to reduce the maximum line loads compared to the “PV+uncontrolled charging” scenario, whereas the district-level optimization shows larger gains. This superiority of the district-level optimization on days without thunderstorms suggests that this approach is also beneficial for dry climates. On days with thunderstorms, however, system-level optimization increases the maximum loads on the majority of the lines (Fig. 2e): Counterintuitively, system-level charging optimization – a

common approach when evaluating V2G effectiveness – exacerbates grid loading in tropical climates. District-level EV charging optimization, in contrast, is able to substantially reduce the loads compared to the combination of PV with uncontrolled EV charging (Fig. 2f), yielding reductions of $18.1 \pm 11.8\%$ (mean $\pm$ standard deviation). This confirms that decentralized EV charging optimization, balancing out demand fluctuations at the district level, effectively reduces the burden on the grid due to PV integration in tropical climates. More detailed statistics

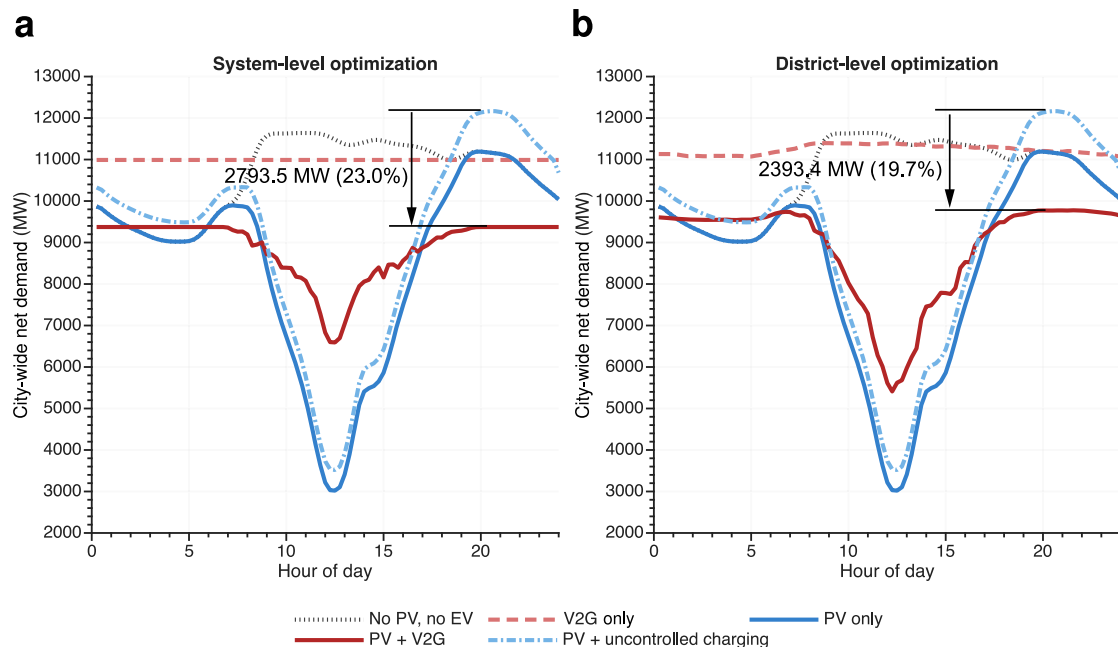


Fig. 3 | District-level electric vehicle charging flattens the city-wide duck curve.
a City-wide net demand profile with system-level EV charging optimization for a typical day without thunderstorms (solar irradiance data from March 22, 2024).
b Corresponding city-wide demand with district-level EV charging optimization on the same day. The arrows depict the effect of V2G on peak net demand reduction

relative to scenarios with PV combined with uncontrolled EV charging. The achieved peak reduction of the system-level optimization is similar to the district-level optimization. EV stands for electric vehicle, V2G stands for vehicle-to-grid and PV stands for photovoltaic.

across all PV and EV integration scenarios are given in Supplementary Note 8. An extensive sensitivity study shows that district-level V2G optimization remains effective across a wide range of different operating conditions, including varying battery capacities, (dis)charging rates, EV adoption levels, and PV integration scenarios (Supplementary Note 9). Moreover, a detailed alternating-current (AC) power flow analysis shows that network voltage levels remain within acceptable limits (Supplementary Note 10). Generally, we find qualitatively similar, but less strong, effects when using uni- instead of bidirectional EV charging, i.e., not allowing EVs to feed energy back to the grid (Supplementary Note 11).

Is district-level “PV+V2G” also able to reduce the original duck-curve problem at the aggregate, city-wide level? Indeed, as shown in Fig. 3, the large-scale deployment of PV (“PV only”) induces a strong dip in the city-wide net demand in the early afternoon, especially on sunny days (for days with thunderstorms see Supplementary Note 6). System-level “PV+V2G” almost completely flattens this duck curve (Fig. 3a). District-level “PV+V2G” is able to reduce the peak to a similar degree, while some dip in the demand curve persists (Fig. 3b). However, the demand increase in the late afternoon is substantially less steep compared to the “PV only” scenario and comparable to the usual base demand increase (“No PV, no EV” scenario) in the morning hours. Thus, district-level EV charging also significantly reduces the original duck-curve problem, albeit to a slightly lesser extent than the system-level approach. Note that controlled charging without PV (“V2G only”) flattens the curve (Fig. 3), but there is no appreciable reduction in the peak demand compared to the base scenario (“No PV, no EV”). Similarly, integrating PV alone (“PV only”) has no effect on the peak demand when only PV is integrated. In other words, only the combined deployment of PV and EVs can effectively lower peak demand and reduce dependence on non-PV generators.

Impact of mobility patterns on V2G potential

Since EVs can only charge or discharge while parked, their spatio-temporal energy storage potential is strongly determined by their

detailed mobility patterns. This is particularly evident during thunderstorms, when solar irradiance drops and recovers rapidly, leading to sudden reductions and increases in PV generation. These generation losses can be compensated through V2G support from EVs, but only if a sufficient number of EV batteries are available – this availability depends on the number of parked EVs connected to the grid.

To explore the importance of this factor, we compare the effectiveness of controlled EV charging on weekdays with weekends. Figure 4a, b illustrates the average number of parked EVs simulated over a month, broken down into weekdays and weekends, in a typical peripheral residential district and a typical central commercial district, respectively. The residential area exhibits an inverted bell curve in EV availability, with most vehicles departing in the morning and returning in the evening, limiting the potential for daytime charging and energy storage. On the weekend, this curve is substantially flatter with many more cars parked during the daytime, which increases the potential of V2G to smooth out local drops in the PV outputs due to thunderstorms. This trend is primarily due to the absence of work obligations for many residents on weekends, reducing the need to commute.

Conversely, the commercial district with its offices and shopping malls shows the opposite behavior with a high EV availability during the daytime and low availability during the night. This pattern aligns with business hours, as many vehicles remain parked at workplaces and commercial places, creating an opportunity to synchronize EV charging with solar PV generation effectively. Furthermore, and again being in contrast to the residential area, the commercial area has a lower number of cars parked during the weekend compared to the weekdays, which reduces the potential of V2G to smooth out local drops in the PV outputs due to thunderstorms. This can again be explained by the absence of work commutes during the weekends.

Based on this observation that EVs tend to be more evenly distributed across the city during weekends, we hypothesize that controlled EV charging reduces grid loadings more effectively on the weekends compared to weekdays. This expectation is confirmed in Fig. 4c and Supplementary Fig. 25, showing the results from the

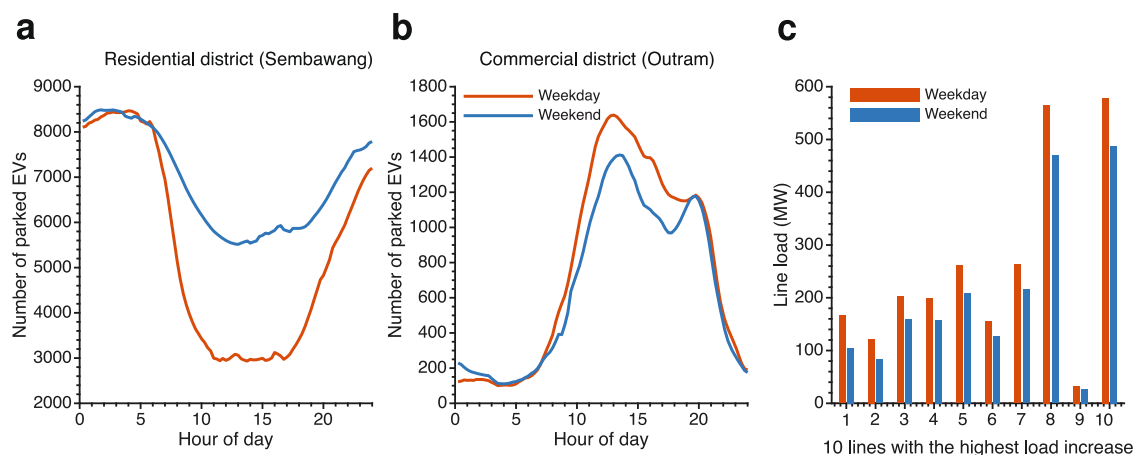


Fig. 4 | Urban mobility patterns affect the effectiveness of vehicle-to-grid. a, b Number of parked EVs in a residential district (a) and in a commercial district (b) during weekdays and weekends, averaged over a simulated period of one month. **c** Comparison of maximum line loads between weekdays and weekends in the “PV

+V2G” scenario with district-level optimization over the one-month period. For the daily solar irradiance, a typical day with thunderstorms is taken (April 20, 2024, same as in Fig. 1a). EV stands for electric vehicle, PV stands for photovoltaic and V2G stands for vehicle-to-grid.

district-level “PV+V2G” strategy. The maximum line loads are appreciably reduced during weekends compared to weekdays, highlighting the importance of considering individual EV mobility patterns for evaluating their impact on the power grid performance. Note that the base electricity demands (i.e., the district-level demands without EVs) are assumed to be the same on weekdays and weekends (Methods), so that these differences are a direct causal result of the mobility patterns.

Discussion

In this work, we have examined the potential of EVs to support the large-scale PV deployment in tropical cities by combining detailed mobility patterns and EV charging optimization with fine-grained PV output data and electricity grid simulations. Using Singapore as a case study, our key insights are as follows. First, rapidly passing thunderstorms, characteristic of tropical climate zones, lead to highly localized drops in PV output, resulting in strong net demand fluctuations and grid overloads. Second, contrary to expectations, controlled bidirectional EV charging aimed at balancing out demand fluctuations at the city-wide level exacerbates line loadings. Third, to overcome this challenge, we have introduced a district-level EV charging scheme that smooths the net demand fluctuations of each urban district and is able to significantly reduce the burden on the electricity grid. Fourth, we have shown that the detailed mobility patterns have a significant impact on the effectiveness of controlled bidirectional EV charging with respect to supporting the integration of PV in tropical climates. In particular, we observe that in some residential areas, EV charging may be insufficient to flatten net demand on weekdays due to limited availability of EVs during the daytime, so additional stationary energy storage systems may be needed to maximize PV use.

Electricity network reinforcements require substantial capital investments and lengthy planning processes, especially in dense urban areas³⁸. The introduced district-level V2G scheme mitigates both the duck curve problem and this infrastructure challenge. Drawing on the literature on the economics of electricity network planning, we estimate for Singapore potential savings of several hundred million up to several billion SGD (see Supplementary Note 13). These main benefits of the proposed V2G scheme must be considered in relation to the associated costs. For EV users, these costs are primarily due to accelerated EV battery degradation and investment in bidirectional chargers³⁹. Using a detailed battery degradation simulation and an economic analysis (details in Supplementary Note 14), we estimate that the required compensation cost to break even with battery degradation and charger costs is SGD 11.20–24.16 per MWh of V2G energy flow.

These values are comparable to those previously reported for low-usage V2G schemes and in the range of incentive payments from current demand-side flexibility programs. The district-level V2G scheme also requires communication and control infrastructure. Although detailed cost estimates are currently difficult to quantify⁴⁰, these components are similarly essential in other V2G systems⁴¹, and we therefore expect their costs to be within a comparable range. Several large-scale V2G pilot projects worldwide have already demonstrated the practical feasibility of implementing the required communication and control systems (Supplementary Note 1).

Our framework can support progress towards policy-driven sustainability and energy security goals. In Singapore, for instance, it may directly contribute to achieving the national net-zero emissions target for 2050⁴² by accelerating the large-scale integration of PV (through deferring costly grid reinforcements) and by enhancing the attractiveness of EVs (e.g., by enabling new revenue streams through controlled charging). Indeed, a recent system dynamics model estimates that net carbon emission savings can reach up to 29.1 million tonnes by 2040, assuming a supportive policy scenario that enables the PV capacity to reach $\approx 8\%$ of the total electricity generation by the same year³². Existing policy recommendations towards this goal could be extended to align more closely with our approach. For example, while Singapore’s national ‘Green Plan 2030’^{32,34} outlines specific interim milestones towards the net-zero target, including a full transition to cleaner-energy vehicles by 2040, our results suggest the additional design of complementary incentive programs, such as attractive compensation schemes for EV users participating in controlled charging. A more detailed discussion of the alignment of our work with Singapore’s Green Plan 2030 and beyond is given in Supplementary Note 15.

Overall, our study highlights that in Singapore, a region with low vehicle ownership, an appropriate EV charging strategy can significantly support large-scale PV deployment. Consequently, we expect that the revealed benefits of the district-level EV charging also apply to other urban regions with similar or higher EV adoption rates, as our sensitivity analysis shows that higher EV shares are associated with stronger line load reductions (Supplementary Note 9). At the same time, a higher number of EVs distributes V2G participation across more vehicles, reducing the average battery degradation per vehicle. Urban structures that differ from the compact spatial organization of Singapore may imply changes in the detailed energy consumption and mobility patterns⁴³. We expect that such changes may have ‘second-order’ effects on the grid load mitigation potential, comparable to the

effect of different mobility patterns during weekends and weekdays (Fig. 4), which should be evaluated on a case-by-case basis. Furthermore, we have shown that our decentralized load-balancing framework can also be beneficial for reducing grid loads in non-tropical settings.

Our work has several limitations that open promising avenues for future research. First, by focusing on urban districts, our analysis may overlook opportunities that emerge at finer spatial resolutions. Future studies could further leverage detailed mobility, PV generation, and local electricity network data to optimize EV charging within neighborhoods or individual carparks. Such optimization would allow users to gain direct monetary benefits through PV and V2G and it would align well with concepts of solar energy communities^{44,45}, thereby further supporting long-term resilience goals such as those outlined in the Singapore Green Plan 2030 (Supplementary Note 15).

Second, a potential limitation arises from privacy constraints associated with mobile phone data and the resulting challenges of accessing such data in other urban regions. The current framework only requires data that is spatially aggregated at the level of urban districts (typical area > 1 km²). Temporally, the data could also be limited to relatively short driving profiles (i.e., each few days, a new random sample of anonymized users can be selected). This allows for a strong data pre-aggregation by the data provider, which reduces the risks of data misuse, ensures compliance with local and international data protection regulations and thus facilitates data access. Moreover, our framework is not limited to mobile phone data but can be applied to any type of mobility data, such as that generated from agent-based models. Recent advances in transfer learning also enable pre-trained deep learning mobility models to be fine-tuned for specific cities using only small amounts of local data (e.g., from volunteer-based efforts)⁴⁶, making them suitable for more fine-grained analyses.

Third, our optimization framework currently omits additional objectives, such as extending EV battery lifetimes through degradation-aware charging strategies⁴⁷ that promise to further increase the economic viability of the proposed V2G scheme. Fourth, behavioral factors, such as users' willingness to participate in V2G programs, preferences for charging speed, and battery state-of-charge thresholds^{48–50}, are not yet considered. Incorporating these factors would further enhance the accuracy of the results. Finally, it remains an open challenge to design effective electricity market models and incentive policies^{51,52} that encourage EV charging behaviors consistent with the patterns assumed in this work.

In conclusion, our work demonstrates that coordinated bidirectional charging of EVs facilitates the timely, large-scale integration of PV in tropical cities, paving the way for effective implementation that supports the transition to a low-carbon urban future.

Methods

This study involved secondary analysis of previously collected, anonymized mobility data and did not constitute human subjects research. Institutional review board approval was therefore not required. Details of the mobility data and their use in the EV mobility modeling are provided in Supplementary Notes 2 and 3.

Base electricity demand modeling

The base electricity demand (not considering PV and EV integration) was modeled using the bottom-up approach in ref. 53. The electricity demand for each building was thereby simulated using the open-source software 'City Energy Analyst'⁵⁴. This software determines demand profiles based on building attributes (building use type, building height, floor area, etc.), weather input, and the surrounding environment. The simulated household energy consumption was calibrated against district-level monthly data⁵⁵, and sectoral energy estimates were adjusted using annual consumption data for each sector from the same source. Subsequently, we summed the

electricity demands of all buildings and then used half-hourly system demand data⁵⁶ for 2022, averaged over all days, to calibrate the simulated energy demand profiles. Next, the future demand profile for the year 2050 was obtained by scaling the peak demand to 11.5 GW, as projected by the Solar Energy Research Institute of Singapore³³, and assuming that the relative half-hourly variations remain unchanged (Supplementary Note 4). In other words, the same base electricity demand curve was used for all days in our simulation (e.g., we did not differentiate between weekdays and weekends) to better understand the impact of PV and EV mobility dynamics. Finally, to get the district-level half-hourly demand profile (for all sectors), we aggregated the calibrated demands for all buildings in a district.

Solar PV generation modeling

Due to the limited availability of real-world PV generation data at the level of the analyzed integration scenarios, a conventional simulation-based method was adopted. Specifically, solar PV generation was simulated for each urban district by combining spatiotemporal irradiance data with the generation capacity of available surface areas. Although indirect, such irradiance-based methods have been frequently used in prior studies and have shown reliable performance in urban-scale PV potential assessments^{33,57}. Building-integrated PV (BIPV) constitutes the largest share of Singapore's overall PV potential due to the high building density. The potential generation capacity of BIPV at the district level was taken from a prior study⁵³, whereas a minimum irradiation threshold of 750 kWh m⁻² year⁻¹ was used to select available surfaces for PV deployment. Building geometries were assumed to remain unchanged for the year 2050. In addition to buildings, our study considers other potential PV installation areas, including water bodies for floating PV (usable surface ~ 0.28 × area of water bodies), and designated land areas for ground-mounted PV³³. For both categories, system-level efficiencies were assumed to be the same as those adopted for rooftop PV⁵³. Finally, to match Singapore's projected PV deployment target³³ of 8.6 GW_p by 2050, the remaining generation was attributed to infrastructure-integrated PV (e.g., noise barriers, PV structures above roads). Aggregated across all districts, this yields a contribution to the installed PV capacity of 9.3% (0.8 GW_p), consistent with the city-wide estimate of 12.6% in Singapore's PV Roadmap³³ (which provides only an aggregate figure). Then, the PV generation profile of each district was derived based on the mean district-level solar irradiance profile⁵⁸, together with the estimated installed PV capacity. A simplified linear irradiance-output relationship⁵⁹ was assumed to convert irradiance into generation. While we note that a non-linear model may offer higher accuracy at higher spatial resolutions (e.g., individual buildings), the linear approximation is appropriate for the current district-level analysis, where localized variations (e.g., local shading and performance ratios) can cancel out.

Uncontrolled EV charging

Charging demand was simulated for an entire month, after which the daily average was calculated to represent typical daily charging patterns⁶⁰. To explore the maximum possible utilization of EVs as mobile storage units, all vehicles were assumed to be connected to charging stations whenever the parking duration exceeds a pre-determined threshold of 30 minutes^{61,62}. A vehicle was then charged if its battery state-of-charge (SOC) was below a pre-defined threshold of 20% or if the current SOC was insufficient to cover the upcoming trips^{63,64}. A backward calculation was used to determine the minimum SOC required to satisfy subsequent trips at each time step, starting from the last trip and working backward. This calculation accounts for the maximum allowable charging opportunity (i.e., when an EV is parked longer than 30 mins) and the SOC limits of each vehicle i ($\text{SOC}_i^{\min} \leq \text{SOC}_i(t) \leq \text{SOC}_i^{\max}$, with $\text{SOC}_i^{\min} = 0$ and $\text{SOC}_i^{\max} = 1$).

Through this backward iteration, the minimum SOC required for the next trip(s) at each step was determined. All EVs were assumed to be equipped with an average battery size of 71 kWh^{24,65–67}. We also assumed a slow charging rate of 7.2 kW, which aligns with a typical AC home charging infrastructure⁶⁶. The detailed simulation process is summarized in Supplementary Note 7. The robustness of our results against variations of these parameters is presented in Supplementary Note 9. All simulations were implemented in MATLAB R2023b (MathWorks).

System-level EV charging optimization

System-level (i.e., city-scale) EV charging optimization focuses on the benefits for the aggregate power generation needs by flattening the daily demand curve and thus reducing the required installed capacity of non-PV generators that constantly ramp up and down, allowing for steady-state operation with optimal efficiency. For the EV charging optimization, a day was discretized into $\Delta t = 0.25$ hour = 15 min intervals such that each day starts at $t = 1$ and ends at $t = 96$. The EV charging and discharging states were determined as follows. Consider a power system in which a total amount of N EVs schedule their charging profiles over T time slots, each with a duration of Δt . Let $O(t)$ and $S(t)$ denote the base (“No PV, no EV”) electricity demand and the solar PV generation at time t , respectively. The net non-EV demand is $D(t) = O(t) - S(t)$. To flatten the aggregate demand profile seen by the grid, the quadratic deviation of the total load is minimized²⁸:

$$\min_{p_i^{\text{ch}}(t), p_i^{\text{dis}}(t)} \sum_{t=1}^T \left(D(t) + \sum_{i=1}^N [p_i^{\text{ch}}(t) - p_i^{\text{dis}}(t)] \right)^2 \quad (1)$$

$$\text{s.t. } 0 \leq p_i^{\text{ch}}(t) \leq \bar{p}_i^{\text{ch}}, \quad \forall i, t \quad (2)$$

$$0 \leq p_i^{\text{dis}}(t) \leq \bar{p}_i^{\text{dis}}, \quad \forall i, t \quad (3)$$

$$\text{SOC}_i(t+1) = \text{SOC}_i(t) + \eta_i^{\text{ch}} p_i^{\text{ch}}(t) \frac{\Delta t}{c_i} - \frac{1}{\eta_i^{\text{dis}}} p_i^{\text{dis}}(t) \frac{\Delta t}{c_i}, \quad \forall i, t \quad (4)$$

$$\text{SOC}_i^{\min} \leq \text{SOC}_i(t) \leq \text{SOC}_i^{\max}, \quad \forall i, t \quad (5)$$

$$\text{SOC}_i(t_j^{\text{dep}}) \geq \text{SOC}_i^{\text{req}}(t_j^{\text{dep}}), \quad \forall i, j \quad (6)$$

where $p_i^{\text{ch/dis}}(t)$, $\bar{p}_i^{\text{ch/dis}}$ and $\eta_i^{\text{ch/dis}}$ denote the charging / discharging power of vehicle i at time step t , its maximum value and the corresponding efficiency. The battery capacity is denoted by c_i . The j -th departure event of vehicle i is given by t_j^{dep} , and $\text{SOC}_i^{\text{req}}(t_j^{\text{dep}})$ is the minimum state-of-charge required to complete the subsequent trip(s) starting at that departure time (pre-computed as in the uncontrolled charging case). We assumed a fixed charging and discharging power for all EVs, equal to the charging rate used for uncontrolled charging. The efficiencies were set as⁶⁸ $\eta_i^{\text{ch}} = 0.87$ and $\eta_i^{\text{dis}} = 0.90$. The EV charging optimization problem was solved using Gurobi Optimizer (version 12.0.0, Gurobi Optimization, LLC).

District-level EV charging optimization

Conversely, this optimization emphasizes more local benefits by minimizing the maximum net demand within each urban district, ensuring a more balanced distribution of energy demand across

urban areas:

$$\min_{p_i^{\text{ch}}(t), p_i^{\text{dis}}(t), z_k} \sum_{k=1}^K \frac{z_k}{D_k^{\max}} \quad (7)$$

$$\text{s.t. } z_k \geq D_k(t) + \sum_{i \in I_k} [p_i^{\text{ch}}(t) - p_i^{\text{dis}}(t)], \quad \forall k, t \quad (8)$$

$$\text{Eqs. (2) – (6)} \quad (9)$$

where k represents the index of districts and z_k represents the peak demand of district k . I_k denotes the set of vehicle indices within district k and K is the total number of districts. $D_k^{\max}$ is the peak base demand (without PV and EV integration), which was used to normalize the electricity demand, given that its magnitude varies greatly across the districts.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Electricity demand and solar irradiance data are publicly available from the Energy Market Authority of Singapore (<https://www.ema.gov.sg/resources/statistics>). Data for the derivation of location-specific PV generation potentials are also publicly available, as described in ref. 53. Travel survey and census data are publicly available from the Singapore Department of Statistics (<https://www.singstat.gov.sg/publications/reference/cop2020>). Although raw mobility data are not publicly available due to privacy considerations, district-level aggregated data to reproduce the findings and figures in this paper are available at <https://github.com/tropicalcityv2g-hash/Powering-tropical-cities> and have been archived on Zenodo⁶⁹.

Code availability

All code used in this study is publicly available at <https://github.com/tropicalcityv2g-hash/Powering-tropical-cities> and has been archived on Zenodo⁶⁹.

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Acknowledgements

M.S. and H.Y. acknowledge support from the start-up funds provided by Columbia University. Part of this research was conducted at the Singapore-ETH Centre, which is supported and funded by the National Research Foundation and ETH Zurich, with contributions from the National University of Singapore, Nanyang Technological University and the Singapore University of Technology and Design.

Author contributions

J.Z. developed the models, analyzed the empirical data, conducted the numerical simulations, and wrote the manuscript. T.D. contributed to the data analysis and numerical simulations. H.Y. contributed to the power systems modeling. S.Y. and B.-S. L. contributed to the discussion of the results and reviewed the manuscript. M.S. conceived the project, designed the study, supervised the work, and was the lead writer of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-026-71123-6>.

Correspondence and requests for materials should be addressed to Markus Schl pfer.

Peer review information : *Nature Communications* thanks Xingxing Zhang and the other anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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